

Video Transcripts

Module 1: Introduction to the Artificial Intelligence Design Process

Video 1: Introduction to the AI Design Process (03:01)

In this module, we will learn about four activities required to design an AI product. We will describe each of these four activities and illustrate their importance in the design process by using examples of different AI technologies with a focus on Natural Language Processing. Before we dive in, let me give you two reasons why the design process in AI is so important.

The first reason is because of the rapid evolution of AI possibilities. This means that finding practical applications almost always requires some form of innovation. Indeed, it is very hard to copy an AI application, and they are almost always unique. This is very much unlike in other areas where you can do as your neighbor and practically get the same car, clothes, phone, tennis racket, education, and even house. In AI, copying what others do without some design work will probably result in sub-optimal outcomes.

A second reason, co-founding with the first, is the importance of AI complementary assets. Very often it is not that AI in itself provides value. What does provide value is the combination of AI with changes in business processes. In some cases where you need a piece of AI that does not exist such as a predictive model unique to your company or a specific use of computer vision to distinguish yourself from the competition, one has to design a deployment process where the organization will keep improving the technology by feeding the right training data to the machine learning algorithms and at the same time will be adapting their business processes and products to benefit from such gradual improvements. Thus, one has to design AI products with a vision of how the organization will adapt the technologies deployed and improve.

I suggest that before you continue, you stop for a few minutes and identify some AI products you would like to design. For each of these design ambitions, keep a log throughout the course where you identify how "what" you're learning relates to each of them. By the end of this module, you will be able to organize the design effort into four activities. By the end of the course, the objective is to prepare you for many successful AI product design efforts.

Video 2: The First Stage of the AI Design Process (07:04)

We will now review the four stages of the design process, starting with the first one. I will use a certain logic to explain the stages one after the other, to help you understand what is needed to design an AI product. However, each stage is an activity in itself. And all can really be done simultaneously or in a different order than the one I will present them in.

In this first stage, we have to identify a behavior we expect artificial intelligence will do. Basically, the AI without the "artificial", i.e., let's just think about intelligence. The intelligence that we will aim to use eventually in an artificial form. There are two challenging choices that need to be resolved in this first stage. Performance metrics and scope.

I will now introduce these two challenging choices and show some examples on how to address them. I'd say one of the biggest challenges, if not the biggest challenge in the design process is to clearly spell out what are the target intelligence performance metrics that we want to implement with artificial intelligence? And this is difficult because what AI can do is so rapidly evolving. That is, the performance frontier in our chosen domain is most likely evolving rapidly.

An easy illustration of the speed of change of the performance frontier can be seen very clearly in what happened with computer vision in the first decade of the century. You will see on this chart how that recognition rate on a big dataset of images called ImageNet evolved over time. For benchmarking purposes, the human error rate on this classification task is 5% because the images are challenging and difficult. This means that if your choice is to set yourself a performance metric of 5%, then you want to implement an AI that is as good as humans in this task.

And in only five years, as you can see in the chart, from 2010 to 2015, recognition rates went from very bad to superhuman performance. In 2010, the state of the art in artificial intelligence had an error rate of 25%. That is, one in four images in the dataset were incorrectly labeled, far away from the 5% human error rate. However, in the subsequent five years, AI improved so rapidly, that it surpassed human performance. That is what we call superior human behavior. And imagine what this five-year evolution meant for business models based on computer vision that started in 2010. They had to forecast this evolution and take it into account into their plan. A second and related challenging choice, the second one of this first stage, is to spell out what is the scope of this superior behavior that we are expecting to implement?

For example, in the case of self-driving cars, these classification tasks may give us a hint that it's possible to locate all traffic signals in the road just with computer vision. And, in fact, those performance levels achieved beyond 2015 are consistent with the rapid advancements of self-driving cars and the associated technologies. But there may be unforeseen challenges between existing benchmarks like ImageNet, and what we're trying to achieve in our target domain like self-driving cars.

In fact, there are a number of differences between the ImageNet classification task and what self-driving cars have to do such as the fact that self-driving cars process video sequences and not static images, and therefore have more information, and more information to process. Or the fact that some traffic signals may be hidden. Or the error rate that may be tolerated is definitely not 5%.

So, in this first stage, the more precise you can be for what is important that the artificial intelligence does, the better. But it's okay not to be extremely specific until you've gone through the next stages at least once. Especially if this is the first time, you're thinking about it. In some cases, like in the self-driving case, the models chosen, and the approaches chosen, the selections done may restrict what the self-driving option is implemented. Instead of traffic signals, you may choose to focus on finding pedestrians or bicycles. And then you may decide that to distinguish between them and inanimate objects, it's a good idea to use infrared cameras that help you identify warm objects like people. And in some cases, those warm objects or the infrared signals are a complement that can work together with computer vision, iron out some of the issues, and help you bring down that 5% error rate on the ImageNet case for human.

Therefore, what we need to do in this first stage is to address these two challenging choices. First, the performance metric choice. Understand how the performance frontier may be evolving in our chosen domain, so we can send targets for the AI design. And second, the scope challenge. Understand what is the scope of what we're asking AI to achieve. What is the intelligence we need in our application? The four stages in the design process may have to be revisited more than once as I hinted before.

So, if this is the beginning of your design process, it's okay to be a bit generic and come back after you've done a first iteration of stages two, three, and four. The second time around you may be better equipped to address these two choices, metrics, and scope. It is also possible that the

second time around you pivot your original thinking based on finding from your first iteration, like many self-driving companies did, lowering their standards into finding specific solutions like detecting pedestrians or bicycles sometimes in restricted environments, like in urban environments where many of the accidents occur.

Video 3: The Progress and Potential of Natural Language Processing (NLP) (06:02)

Let's now move away from computer vision and explore as an example, another area that is also experiencing a lot of progress, natural language processing. The artificial intelligence that we may want to include in our design may be related to one or more natural language tasks, as it was with computer vision. In the specific case of natural language, there are many tasks we may be interested in, such as translating product documentation, annotating news, describing words in a given scene or data spreadsheet, summarizing a long document into a few sentences, filling forms of specific customer requests, etc.

The field of natural language has seen a lot of progress towards some of these and other tasks. Some basic required functions such as name, entity recognition, or positional tagging are practically solved at a level beyond human performance, and we are reaching human-level performance in many other of these tasks. And even better, there is excellent progress also in more elaborate tasks such as the sentiment analysis or core reference resolution, identifying expressions, referring to a given entity in a text or word-sense disambiguation, and parsing, and many others. The combination of all these progress has made services like machine translation progress immensely.

The state-of-the-art in NLP includes advances such as transformers which can have very general-purpose applications. An example of these is that transformers produced by Open AI and its competitors. The first transformer to hit the market with great success was GPT-3, which in its time was the biggest investment example in terms of effort and design and learning time. It was trained basically almost on all the written text available. Think about it, anything that humans have written was given to the system. It became the largest model in terms of parameters, with over 50 billion of them, that's more than inhabitants on the planet. It also required more than \$5 million just on electricity costs to power the computers while training the model. Development since then have made all these astronomical figures seem small as companies race to build the largest possible infrastructures to train their models.

You'll get a sense of what transformers can do when you work through some examples. What is really hard for transformers, the technology behind large language models, are tasks that involve applying common logic of the sort, a 5-year-old will be able to do, or holding really long coherent conversations. There are ways around many of such hurdles, as we will see, that involve things like retrieval-augmented generation, chain-of-thought, and tools. What's very exciting about these additions to transformers is how they are pushing the envelope of agentic AI to the point where transformers are no longer one shot companions, but real, independent agents that have the potential to autonomously do great things for us. The combination of all of the above enables natural language applications to do tasks such as purchase your plane tickets, make hotel reservations, or send flowers to your loved ones. They can even act as a college companion that attends class for you and make sure you learn everything that is required to pass the exam. These last application would require some human supervision, given that transformers still hallucinate. In fact, hallucinations such as when a transformer invents a statement in references of fake sources seem very hard to eradicate completely. That's still very difficult in general, and even from a theoretical point of view. Any application thus must make sure that hallucinations are either not a serious problem or not present at all. Later in the program, you will do a deep dive into transformers. This will include reviewing the technical architecture of a transformer and how it impacts the four stages of the AI design process. You will also cover retrieval-augmented generation, chain-of-thought, and tools, and more broadly, the fundamentals of agentic AI. Think of agentic AI as the "opera conductor" that can autonomously take digital and AI components and make them work collaboratively to create ever more complex AI products and services. Let's now move to the second stage and shift the focus to business fundamentals.

Video 4: The Second Stage of the AI Design Process Part 1 (09:54)

Let's now talk about the second stage in which you identify the business process that AI will support. To do this, there are two key issues that need to be addressed. First is the strategic one, deciding how AI will play out to your long-term sustainable advantage. Second is the operational one that involves specifying the business process where AI will intervene in setting performance targets for such processes, not just for AI, but also for its required complementary assets. Let's start with the first of the two, the strategic AI choice. Here you need to decide how you will use AI to compete in the marketplace. That is, what is the role you want AI to play in your design process?

One way to think about this is to use the Delta model, developed by Professor Hax at MIT Sloan, which postulates that there are three ways to compete. Let's now review the three of them. The first strategy is to be a best product player. In the case of AI, this means having the best technology possible. For example, the best fingerprint recognition software. Anyone interested in applications that are based on fingerprint recognition will want to use your product if they can afford it. Another strategy, the second one in the Delta model, is to be a full customer solutions player. In the case of AI, this means providing a product that incorporates all AI and all the associated features that will make it a useful product or service. For example, a security and access control solution for buildings that maybe your company already has and now incorporates fingerprint recognition software in other functionalities required by customers, such as locks, intrusion alert mechanisms, connections with the police, 24/7 recovery services, mobile app unlocking, etc.

You may not have the best fingerprint AI in the world, but you're offering a service that is useful to a certain segment of customers that, therefore, will buy it. A third strategy in the last one of the Delta model is network externalities. Here the focus is on building the largest user base possible in a way that results in users getting more benefits as the size of the user base increases. For example, you may focus on building a nationwide database of fingerprints so that police can quickly identify people in criminal scenes. The more municipalities and states that participate, the more likely it is that the database is useful. You may need a AI for fingerprint recognition, but you don't need to have the best one and you don't need to provide a full solution to the police. Most efforts involving self-driving cars, like the one we discussed when talking about the first stage of the design process are focused on the second strategy in the Delta model, full customer solutions. This is because the use of AI is closely related to the design of the full product. In fact, most automobile brands have different flavors of how they use AI to support intelligent driving features and all of them give a complete solution to the user.

Before moving to the second choice in the second stage, let's explore another computer vision example. Imagine, instead of self-driving cars that you are thinking about offering an automatic image captioning service more elaborate than what is currently available online. Take for example the image on the screen. Current state-of-the-art services can identify specific individuals in images, understand social dynamics, and even generate detailed narratives about what's happening in a scene. However, AI still struggles with deeper contextual understanding

that requires real-world knowledge and cultural nuance. For example, while AI may recognize that someone is making a gesture and that others are laughing, it may miss that this is a specific cultural reference, inside joke, or historical context that gives the moment its true significance.

AI also has difficulty with temporal reasoning, understanding whether a moment captured represents a genuine candid interaction versus a staged photo opportunity, or predicting what may happen next based on subtle social cues that humans intuitively understand. Most importantly, AI has a hard time detecting fake news. It's quite surprising. AI is very good at deceiving itself. Going back to our example image, because this arrangement of characters is not usual, one may naturally question its validity. This means that you will need to address scoping challenges by somehow narrowing the requirements so that the AI can produce results consistent with what you set out to achieve.

In general, when you hit an AI frontier like in this case AI's inability to detect fake news, your creative business juices should start flowing. This is a sign that it's worth digging deeper into the specifics, understanding that this new frontier you've hit will help you in scoping the intelligence that you need to automate. As another option, you may be able to focus on sentiment analysis of the people in the image and create a captioning revision software that makes sure captions are consistent with such sentiment analysis. For example, AI may struggle with nuance, cultural context, or sensitive situations that require human judgement. But if you want a service for a newspaper, perhaps it can extract the relevant information from the associated text to describe the picture with a specific genre and with a small phrase. It may not detect if the image is fake or not, but it may at least provide a description consistent with an associated piece of text. Or maybe you want to translate newspaper picture captions into different languages, which moves you completely away into another dimension in the realm of machine translation that we've also talked briefly about.

In both cases, you may be able to produce the best product in line with the first strategy in the delta model, something that all newspapers can use. The second strategy, a full customer solution, is also possible, for example, by integrating the solution into a specific content management system that maybe your company operates or that your newspaper company has. And finally, the third strategic option may be the best one. You may create network externalities if you can produce an open system that all newspapers use in a way that allows you to capture

how editors modify AI suggestions so you can continuously improve the service. The system may not detect fake images directly, but may benefit from the collective judgment of all users of the system.

Video 5: The Second Stage of the AI Design Process Part 2 (04:43)

Let's now move to the second choice. In the second stage, which is operational. What is the business process that will benefit from AI? As part of answering this question, you will need to also address what are sensible improvement targets for that process and what are things that you need to address within the process, not just in AI. The idea here is to focus on the process that will be impacted by AI so that it can give the engineers adequate information on what is expected of the AI implementation. You need to define a business process and how AI will improve its performance targets. In the next stage, that is, in the third stage, you will be focusing on choosing the specific technology and what AI can be introduced based on the performance targets that you've set.

So, let me give you some examples of operational targets that you can set in the second stage, in the case of natural language processing. If you apply NLP to answering questions in a call center, you may want to lower uncompleted calls by, say, 80%, the same cost as before, so that the responses that the system gives have improved and there are fewer issues with those answers. Another option is that you lower total call center cost by 70% while keeping the same service levels.

As another example, you may want to lower legal cost associated with translation in a given process by 80% by producing a first version of the translation that the lawyers only have to review once, focusing mainly on the tricky places. Or you may want to add voice commands to a web page so that it automatically purchases flight tickets. Very few people talk to web pages. You may want to add voice recognition in the web page so that users can navigate and execute commands verbally, eliminating the need to use a keyboard. It's important to stress that beyond these targets, there are other things you must consider in connection with the AI process you are designing. For example, you may need to see authorization from users, which means a new consent form must be developed, or you may have to renegotiate your insurance terms.

In this stage, you need to define the process well and make sure you address all the required complementary assets. Research by Professor Brynjolfsson at MIT Sloan has shown that

investments in complementary assets can require as much as 90% of the total investment dollars. So they are very, very important, and often we say, they are the bulk of the investment iceberg. We tend to see the AI investments, but most of the real investments are not AI technology related. In summary, what you need to do in the second stage is to address these two challenging choices. First, the strategic decision to use AI as a best product full customer solution or for creating network externalities. And second, the decisions involved in the operational description of the processes involved including the non-AI requirements and specific process performance metrics where AI will help, that is the AI without the AI. Let's now move to the third stage, where we will shift to the choice of AI technologies.

Video 6: The Third Stage of the AI Design Process Part 1 (04:44)

So far, we've made it through stages one and two. In stage one, we focus on two key choices, AI metrics and scope. In stage two, we also focused on two choices, the strategic role AI would play, and the operation targets we would set for the business activities when AI was going to play a role. In other words, before you start with stage three, you have already identified an intelligent behavior you'd like to implement using artificial intelligence, and you also have a business objective that you would like to achieve. The latter includes some business activities where artificial intelligence will be used.

In this third stage, focused on AI technology choices, there are also two key decisions to be made. This is a good mnemotechnic for those that like them. Two choices in each of the first three stages. Well, and what about the fourth one? What do you think? Exactly. Also, in the fourth stage we will have two choices. Four stages, two choices in each. Yes, that's eight choices in total. Learning science tells us that this type of coincidences are great anchors for learning. I hope it helps you retain the content.

As an AI product designer, you have to make eight choices. Eight is also a key number in dancing. Learning science also tells us that if you search on your own why dancing is mostly based on eight step sequences, that will help you retain this important number in AI. Learning science tells us that it must be you making the effort, so I won't reveal anything more about dancing. It's another coincidence that probably will help you. Making connections in your brain between different areas helps you solidify the learning and the versatility across domains. So, let's go back to AI which is really why we're all here.

On the third stage, the first choice of the two involved centers on selecting the right intellectual property approach, and the second one on picking the right data strategy. Let's now review these two choices, IP around the technology and data entered. So, now, starting with IP or intellectual property or the choice of the technology that you'll use for AI, thinking about stages one and two, they have provided you with the right specifications of what artificial intelligence technology will add to your business strategy. Now, you need to be more concrete and pick a specific artificial intelligence technology that you can incorporate in your product.

This selection is very difficult because AI is a general technology, and there are many choices evermore. We're talking about thousands if not tens of thousands. To put the number of technology options in perspective, know that up until 2012, the number of patents related to AI was not very large, and it was not growing significantly. However, since then, the number has skyrocketed, growing year after year, and only in the U.S. there are thousands of patents awarded every year. This makes the life of AI product designers very exciting because there are always new things. But also, very risky, and tricky because one may pick something that is just obsolete in a few months.

Confounding with a growing myriad of choices, there is the lack of off the shelf alternatives. The good news is that there are so many new opportunities for AI that chances are you'll be able to patent what you design. And this should also be part of your IP approach. This course is not going to teach you how to set up a patent, but if your use of AI is novel, useful, and nonobvious, I strongly encourage you to consider patenting. The rest of this course will help you get a better understanding of what these technology choices may be.

Video 7: The Third Stage of the AI Design Process Part 2 — Data Strategy (09:45)

You will learn in the next modules what are the specific advantages and disadvantages of each machine learning approach and what roles AI may play in human-computer interaction, in designing new Superminds, and in manufacturing robotics. What all of them have in common is one thing, data.

In this third stage, the second decision you have to take is what data strategy you're going to follow. In fact, there is a close connection between the data you have and the algorithms you can use with it. The algorithms are related to your IP and the data to the data strategy. And one of the critical issues on data is how it's labeled. This labeling is in fact data about data, and it's why it

is called metadata, and it includes such things, at a minimum, as date, time, collection mechanism, and description of its content. If your data is not properly tagged or if you don't have enough metadata associated with it, you'll be restricted to what you can do.

Perhaps even only unsupervised learning. In many ways, though, the data strategy is also disconnected from the machine learning algorithm. The more data and metadata you can have, the more options you'll have. Perhaps you have data for which no AI algorithm has been already developed, but this may change eventually. This happened, for example, with a picture libraries on the phones. Now that we have face recognition, they are a lot more valuable than we didn't because they can be organized automatically by person. So, you can automatically organize photographic memories by contact. It also happened with text. Most of text written by humankind was never used to AI until recently GPT-3 read everything that was available. And in doing so, it was able to give new use to the collection of text that very few people read.

Who knows? One day AI may read scientific papers and invent new theories and design its own experiments. The opposite feedback force also exists. If you have AI technology that automatically adds metadata to your existing data or that generates new data, you'll have more data upon which to build further AI algorithms. You may even be able to sell it as a business. This is the basis of some of the most successful AI players. Google, in fact, sells your data for advertising purposes. It keeps tagging data and lets others decide how to use it. In fact, the big AI players in the world all have very sound data strategies.

The so-called FAANGs, which stands for Facebook, Apple, Amazon, Netflix, and Google, now Alphabet, have a data strategy that gives them network externalities, allowing them to really test any of the machine learning approaches you'll be exposed to in module two. The data approach of the FAANGs share many things in common, and the end result is an advantage thanks to the data they have.

That's why a lot of people say data is king or data is gold. Let's now see these similarities a bit more in depth. And hopefully, you'll learn from that and get inspired to develop your own data strategy. I'll use a framework, as you can see on the screen, to help you understand the sources of this superior long-term sustainable advantage that data gives. The value from the data, according to this framework, comes from three interrelated forces that fit into each other.

Companies that can make their AI strategy accommodate these three forces will gain a natural long-term sustainable advantage thanks to their data superiority.

Externalities is the first of these three interrelated forces. We've already talked about this when covering stage two as a possible strategy leveraging AI. Take Facebook for example. For a given user, the value of a social network is based on the amount of users the network has. If two networks have similar services, but Facebook has 95% of the "friends," quote-unquote, of a given user, and the other one only 1%, that user is definitely more likely to pivot towards Facebook, everything else being equal, because it is the one that has more of its social circle.

Network externalities is not a concept restricted to AI, and it operates similarly in other domains like with the phone network. What would be the value of a phone if you could not talk to anyone? The more users that have access to a phone, the more valuable it is. In the case of Facebook, this also means that it is the network with the most data related to social networks and therefore the one that can better test artificial intelligence algorithms related to advertising. That explains why Facebook has been so quickly at purchasing other big social networks like WhatsApp or Instagram. For both of these, the possibility of integrating their services with each other meant that the collective value was larger for everyone.

In summary, network externalities has three components that pivot users to the largest of the various options in the market because the larger the community, the more value per user it can deliver. And because the larger the community, the more value the asset has for operators that target the users of the network, such as advertising companies. It is really in the interest of everyone to have one big network where all can join. This is true online where you can go anywhere with one click. It sort of also applies on ground. Who would want two Sunday markets in a small town one kilometer apart from each other, If you could have both of them be one and in the same place? From the point of view of the customers, they could go to a place and compare options instead of having to go back and forth between the two. And for the sellers of goods, they would only need to serve one location with advantages translating to key retail metrics such as out of stocks, shrinkage, operating costs, etcetera.

The second force of the network externalities is system lock-in, and it relates to how the FAANGs increase switching costs from their services. In the case of Facebook, Instagram, and WhatsApp, this means that you have all your contacts, pictures, message history. With groups, they make it

super hard to change because in theory, you'd also need to change all of your contacts and at the same time to have the same experience somewhere else. It also means that you've learned how to use the application, including all of its features. You have set your privacy settings, notifications, profile picture, and other configuration options that make it hard for you to switch.

But system lock-in does not apply only to end users. It applies to the advertisers, developers, and third-party application developers. Perhaps your company has a company license or has pre-approved advertising terms with Facebook, which would require a new negotiation if you were to change. Who knows? Perhaps you have developed your own AI algorithms, like hotel reservation systems do, and that is based on a history of Facebook information. It'd be practically impossible to change all of these without significant costs. That's why it's called system lock-in. You've been locked in, your suppliers and any other provider, too.

Finally, the third interrelated force are economies of scale, which operates alongside system lock-in and network externality. As a result of the other two forces, the winning player in the network externalities race will see its revenue rise. See, for example, the advertisement revenue of Facebook between 2009 and 2019 before the COVID pandemic hit the market. You see that these revenues grew exponentially from less than a billion to tens of billions. What this means is the possibility to afford a larger R and D budget to continue to fund whatever future improvements are needed to increase the size of their user base or even to buy competitor. And more users means more data, and that means better trained AI algorithms.

Video 8: The Fourth Stage of the Design Process Part 1 — Software Development (05:58)

When you arrive at the fourth stage, you already have your AI metrics and scope, your AI strategy and operational targets, and your IP and data approaches. Now it's time for action. You need to get the engineers to work on the AI algorithms, and the rest of the organization to align on the complementary assets. Tinkering with the technology is more effort than any other aspect of a technology, and gives rise to the second iceberg, the tinkering iceberg.

One usually only sees the more glamorous steps. Specification and the key architectural decisions. But the time spent on finding and fixing bugs, and on making the software work in the

production environment is one order of magnitude more effort than anything else in the technical process.

Here, there are again two key choices you'll have to make. The first one has to do with your software development approach, and the second one with how you're going to deal with the AI cancers. The software development strategy has implications similar to key choices made in other types of businesses. Just to give you some intuition, for example, the choice of menu in a restaurant or the location of franchises in a retail chain or the planogram in a grocery store.

What all these choices have in common is that they have a big impact on the business outcomes, and that pivoting choices based on outcomes is important. It's sort of like the fine tuning of the business model. One may test a wine menu in a restaurant and realize that sales are not as large as the ones that same menu has in another similar restaurant in another town.

This may be due to many factors, and it's important to decide whether to pivot the menu or not. In AI, this pivoting is also important, and the software development strategy needs to be such that it can allow it. Reasons to pivot may include a new source of data that increases accuracy or lowers computational costs of the machine learning algorithms or that it is sound that users don't like the tone of the voice responses generated.

This case may be similar to the wine menu above, or perhaps there is a new channel to offer our services, and AI needs to be readapted or we're using a package like TensorFlow version two, and then all of a sudden code moves to a new version or to PyTorch, and we need to revisit all of our code, or many other reasons. Most importantly, flexibility needs to be built into the software development process because, as we said, when talking about stage one, the frontier of AI is rapidly moving, and it is important to incorporate adaptability into the design process. The reason may be mechanical.

Research on robotics at MIT is continuously exploring with new materials. And we are far from achieving human level performance. We don't even have a robot that can insert the key into a keyhole as well as humans can. By way of comparison, humans are born with 300 bones, ending up with 200 when adults. You also have muscles, tendons, ligaments, bursae.

In contrast, robots have at most a few hundred parts. As we build more and more complex mechanical robots, the associated software tinkering will be more sophisticated. This course is

not about software development, but it is critical that you incorporate in your AI design process a basic software development plan. If you don't know how to do it, you can reach out to your technical team and make sure they explain you the basics. The more things we can automate at the development process, the better.

Here I'll name four things that are the minimum you'll have in place before starting. First, a code and data repository methodology. This includes establishing where the different versions of the code will be kept, and what tools to do so will be used. GitHub is one popular possibility that we use at MIT. You will probably need at a minimum a process to transfer code from development to testing to production. Part of the code repository must include documentation so that the code can be reused and executed.

Second, a testing and experimental plan. This means, how are you going to test the software? And how will you manage the different issues that arise from the testing plan? In particular, what are the methods you're going to track, and how are you going to make sure critical issues are addressed? Testing does not mean only software. It may also mean all of the things like business profitability, user acceptance, or even integration with the overall company product strategy.

Third, a GPU experimental set up that rationalizes the use of resources. GPUs or graphic processing units are the computers that do most of the number crunching for machine learning applications. The experimental setup should be such that the use of the available machines is maximized and optimized. If one uses a cloud supplier, it should be in the context of an optimized approach, computationally and financially.

And this brings us to the fourth and last. A plan with a budget to justify the investment. You should know that the rule of thumb is that new software development projects will cost two and a half times more than what you think they'll cost even after applying these rules. They always run over.

Video 9: The Fourth Stage of the AI Design Process Part 2 — AI Cancers (06:55)

Last but definitely not least, the eighth decision, the second of this fourth stage, but the eighth overall in your design process is related to AI cancers and limitations. AI has some known issues that you'll have to plan ahead for. These are things we don't really yet know how to solve very, very well.

Here, I will talk about three of these cancers and how you may deal with them. The first one are adversarial attacks. Research has shown that you can fool any deep learning AI algorithm with simple modifications in the input signal. In the screen, you see an original image of a benign melanocytic mole that when combined with very low intensity adversarial noise, fools the algorithm. One can prove theoretically that with deep learning, this is always possible.

Here is another example where sticker fools an algorithm into thinking that the banana is a toaster. More concerning is that researchers have been able to show that self-driving cars can be made to believe that a stop sign is a speed symbol sign, as you can see on the image. The image on the left does not fool the self-driving car, but the image on the right, does. For the human eye, both cases are clear cases of stop signs, but it's just these little stickers that fool the system. And adversarial attacks can really compromise the security and privacy of the user. Research has recently been able to demonstrate that you can activate and control Siri with commands inaudible for the human ear, and you can make the phone do certain things.

The second cancer is lack of generalization. In many cases, AI cannot generalize properly. This happens, for example, in natural language processing as we sort of indicated earlier. Translation accuracy when going from language A to language B does not help when going from A to C. However, A to B and A to C does help B to C to a certain extent. In general, whether a machine learning algorithm will generalize or not is very hard to predict.

For example, face recognition algorithms run by Microsoft Azure or by Amazon Web Services have tens of millions of faces. Yet, they underperform with certain face colors and makeups because they have not been trained properly on this. Here, the issue is one of balancing datasets which makes generalization very difficult. If you overtrain on this forgotten skin colors and makeup styles, you may then skew the algorithms in another direction. A similar cancer is the one we cover next.

Onto the third cancer, bias. Machine learning algorithms often have the bias of the data that one uses to train them. For example, GPT-2, the precursor to GPT-3, had some fairly worrying biases, as you can see on the screen. The autocompletion was pretty biased because it reflected, you know, a lot of the literature that was input into the system that was old literature when text had very strong biases. Similarly, Amazon had to scrap a recruiting tool because it showed male bias when recruiting engineering talent. This happened probably. We don't really know why, but it

happened probably because males represent over 75% of the engineering workforce in most of the FAANGs. In fact, in all the ones that report the number.

Another related cancer is explainability. One of the big issues with modern deep learning algorithms is that they don't necessarily provide any insight into how they reach a conclusion. For example, you try to train an algorithm to distinguish wolves from dogs, as on the screen. And all the images from wolves have snow, like the one on the screen and none of the ones with dogs have snow. Perhaps because they are urban indoor pictures. It may be unclear how the algorithm is reaching the conclusion that a picture has a wolf or a dog. Perhaps it's simply distinguishing the scene backgrounds and not the foregrounds. Maybe it's easier to find whether there's snow or not. And there is a clear correlation in this example between the two discriminating tasks, wolves versus dogs or snow versus urban scene. But no real causality. In general, your AI algorithm needs to explain how it reach a conclusion. Perhaps deep learning is not your best choice.

The final cancer we'll discuss is unintended behavior. In one case, a robot in a shopping mall fell down the electric stairs and injured some customers. In another natural language example, a GPT-3 chat box installed inside the robot also in a retail environment, when asked by a person if suicide was a good idea, it answered, "Yes, I think you should do it." As another unintended behavior example, third one, the first vital self-driver car accident involved an urban Uber driver that was not paying attention, which, in fact, is probably what people will want to do in these vehicles. Uber probably would have to have had two drivers in the vehicle because of the extra effort to be attentive and not doing anything for very long periods of time.

Think about it. Think it's a lot harder to control a driver than to drive because you have to be most of the time doing nothing. But you have to be very alert. Therefore, your final list of decision needs to consider how you're going to minimize the impact of AI cancer. These five plus others that may emerge.

Video 10: Conclusion (03:43)

Okay. So, we are going to look now at an overview of the AI design process by looking at the four stages and the key decisions that have to be made. The first stage, stage one, is about intelligence. It's about deciding what is the intelligence that you want to incorporate in your AI algorithm. Here there are two key decisions. One key decision is understanding the metrics. How are you going to measure that intelligence that you are trying to achieve? So, it's metrics.

And the second one has to do with the scope. What's the scope of the intelligence that you want to apply? Give some insight into where that intelligence has to excel and where it's not that important. So, stage one. Now, when we move to stage one, we already have an intelligence, but where in the business are we going to use it? Stage two is about really the business activity. And here again there are two key decisions related to AI. What is like the strategic role of AI? So, there's here a strategy role. And then the operational one. Here on strategy, there is a delta model that talks about three possible strategies, network externalities, best product, or full customer solution.

Then stage three is about the actual technology. You're making choices about technology. It's about AI tech. And here we talk about two important decisions that need to be made. One is the IP. What's the actual technical and scientific contribution you will be using, and you will be incorporating? Or maybe you develop your own, and you want to have your own patents or maybe you want to use third party fully available. So, that's one. And the second one is your data approach or your data strategy. And role models will be the FAANGs, you know, Facebook, Apple, Amazon, Netflix, Google.

And then finally the last stage is going to be stage four in your AI design process, and we call it tinkering. That's where you are going to put a lot of emphasis in your team, in your technical team, in actually implementing the software, testing it, and improving it. Two key decisions here will be your software development approach and how you're going to deal with AI limitations or what we call AI cancers.

But there are complementary assets here that will take most of your time, and you need to invest a lot in those. So, it's not just about AI if you're going to do successful implementation. And there's a bit of a full picture here with the eight decisions, four stages.